A Lightweight Mamba Network for the Joint Classification of Hyperspectral and LiDAR Data

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***Abstract*—To address the high computational complexity inherent in the joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data, along with the limitation of existing Mamba-based scanning mechanisms in adequately capturing spatial context, this article proposes a lightweight multimodal Mamba (LWM-Mamba) network, termed LWM-Mamba. The proposed network first employs a shallow feature embedding module to extract initial features from HSI and LiDAR data, thereby mitigating the computational burden arising from spectral redundancy in HSI. Second, a frequency enhancement module (FEM) is designed to augment local and global features as well as fine-grained details, while a spatial compensation mechanism is introduced to preserve feature integrity. Subsequently, a multimodal adaptive Mamba (MA-Mamba) module is constructed. This module first adopts a multipath scanning mechanism that initiates scanning from the central pixel along multiple directions, including rotational and inverse rotational paths, to comprehensively capture spatial dependencies. Furthermore, the module incorporates an adaptive state space model (SSM) to facilitate deep interaction and dynamic integration of cross-modal information. Experimental results demonstrate that the proposed network achieves superior classification performance compared with state-of-the-art methods while significantly reducing computational complexity, thereby validating its efficiency and robustness in HSI–LiDAR joint classification tasks. The source code of LWM-Mamba will be made publicly available at https://github.com/Dilingliao/LWM-Mamba**

***Index Terms*—Classification, frequency enhancement, joint classification, Mamba, scanning mechanism.**

I. Introduction

JOINT classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data, which integrates fine spectral information from hyperspectral imagery with 3-D structural features from LiDAR, holds significant promise for applications such as urban planning [1], precision agriculture [2], and environmental monitoring [3]. Although this multimodal fusion approach can substantially improve ground object recognition accuracy, the inherent differences between the two modalities in terms of data representation and dimensionality continue to pose challenges for the design of classification models that are both efficient and accurate.

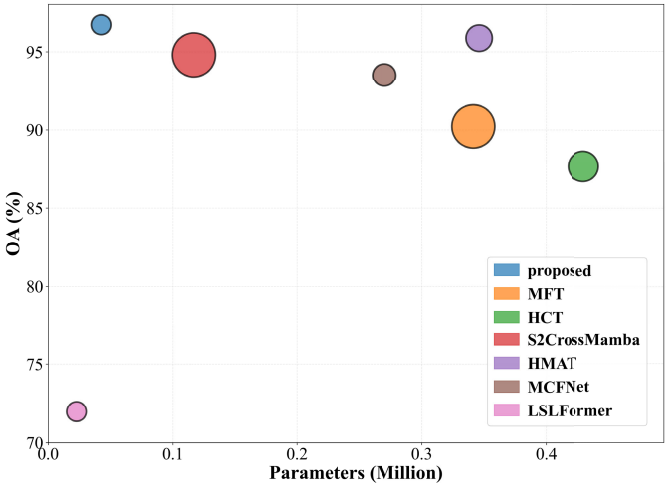


Fig. 1. Relationship among parameters, FLOPs, and OA for different methods. The horizontal axis denotes the number of parameters and the vertical axis denotes OA (%). The size of each circle indicates the FLOPs of the corresponding model, while colors distinguish different methods.

Currently, the task of HSI-LiDAR joint classification is confronted with a significant efficiency dilemma. As illustrated in Fig. 1, a comprehensive analysis of mainstream methods developed in recent years reveals that substantial increases in model parameters and computational complexity do not consistently yield corresponding improvements in overall accuracy (OA). In some cases, complex models are even observed to exhibit diminishing returns, where a considerable rise in computational cost is accompanied by only marginal performance gains. This phenomenon highlights that model lightweighting has become an unavoidable core challenge alongside the pursuit of higher accuracy. Consequently, simply increasing model complexity is not considered a sustainable research direction, and it is imperative that architectures be designed to maintain high performance while operating under low parameter and computational overhead.

In the development of joint classification models, early research primarily relied on convolutional neural networks (CNNs) [4], [5], [6], [7], employing dual-branch or deep architectures to extract features from HSI and LiDAR, respectively. Although these methods achieved satisfactory classification

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performance, they were generally accompanied by high computational and storage costs. For instance, Hang et al. [8] proposed a coupled CNN (CoupledCNN) framework, which addressed the challenge of collaborative multimodal modeling through a heterogeneous feature fusion mechanism. However, this approach suffers from high computational complexity and limited generalization capability in small-sample scenarios. Building on this, to further exploit the complementarity between spectral and spatial information, an effective hierarchical random walk network (HRWN) [9] was developed for the joint classification of HSI and LiDAR data, in which a dual-tunnel CNN structure was employed to capture spectral and spatial features. To address spectral redundancy and modal differences in HSI data, Li et al. [10] introduced a feature reconstruction-guided fusion network. In this method, HSI features are reconstructed via matrix factorization, and LiDAR features are integrated using a cross-modal attention mechanism. Meanwhile, to mitigate the limitations of traditional CNNs in single-scale feature extraction and the efficiency issues caused by excessive parameters, Gao et al. [11] constructed a multiscale residual network (MSRN). Through a multipath structure, features from different receptive fields are integrated, achieving preliminary progress in controlling computational costs; however, the architecture remains relatively complex. Recently, Liu et al. [12] proposed a progressive spatial–spectral-frequency fusion learning (PS2F2L) network, which leverages wavelet transforms to integrate spatial–spectral and spatial–frequency features in a multistage manner, achieving lightweight and effective multimodal classification for HSI and LiDAR data. In summary, although mechanisms such as attention and multiscale processing have been gradually incorporated into CNN-based methods to enhance performance, significant limitations persist due to their inherent local receptive fields and stacked structures, particularly in terms of long-range dependency modeling and lightweight implementation. Beyond pure CNNs, other multimodal remote sensing tasks have also explored physically guided iterative optimization [13] and feature disentanglement [14], yet they still face challenges in joint HSI-LiDAR classification.

To overcome the limitations of CNNs, the Transformer architecture [15], which is based on a self-attention mechanism, has been introduced. Transformers are capable of establishing global associations among all pixel pairs within an image, thereby substantially enhancing a model's ability to comprehend complex contexts. However, this powerful capability is often achieved at the expense of extremely high computational complexity. To fully leverage the advantages of Transformers in joint classification tasks, many studies have combined them with CNNs to balance local feature extraction and global relationship modeling. For instance, a dual-branch approach was proposed [16], in which a novel network based on a hierarchical CNN and Transformer (HCT) was introduced. In another study, Wang et al. [17] designed a convolutional interaction Transformer network. By integrating a Transformer with a convolutional cross-attention mechanism, the fusion of local and global features was improved. However, the computational burden remains substantial. Pan et al. [18] designed a dual-branch CNN-Transformer-based multimodal feature disentanglement-fusion network (DFNet), in which optimization was enhanced through the integration of contrastive learning and knowledge distillation. Although this method addressed the issue of information loss inherent in traditional dual-branch networks, LiDAR features may be weakened in scenarios dominated by spectral information. A graph-attention-based multimodal fusion network (GAMF) was proposed [19], through which the correlation and complementarity between different data sources were explored. Another line of research focuses on multimodal fusion using attention-enhanced Transformers. Zhu et al. [20] introduced a hierarchical multimodal feature aggregation-based multihead axial attention Transformer (HMAT), which consists of a hierarchical multimodal feature aggregation (HMFA) module and a multihead axial attention (MHAA) mechanism. While this approach addressed the problem of insufficient cross-level feature interaction in multisource data, it demands high computational resources. To reduce complexity, Roy et al. [21] proposed a novel multimodal fusion Transformer (MFT) network. Furthermore, to enhance information exchange across different modalities, cross-hyperspectral and LiDAR (Cross-HL) attention [22] has extended traditional self-attention mechanisms. In [23], a novel multiscale cross-domain fusion network (MCFNet) was proposed to integrate complementary information from HSI and LiDAR, while the multiscale cross-attention fusion network (MCAFNet) [24] further investigates cross-modal adaptive fusion for multisource remote sensing classification. Although these methods have achieved significant progress in classification accuracy, the challenges posed by their computational overhead and memory consumption cannot be overlooked. To address this, a lightweight spectral-LiDAR fusion network, termed LSLFormer, was proposed [25]. Although this method addresses the issues of high computational cost and complex cross-modal interaction mechanisms in existing models, its performance remains limited.

To alleviate the high computational complexity associated with Transformer models, the Mamba architecture [26], which is based on a structured state space model (SSM), has demonstrated significant potential. By employing a selective state mechanism, hidden states are dynamically updated according to the input content, enabling excellent contextual modeling capabilities while maintaining linear computational complexity. This provides a novel approach to overcoming the aforementioned computational bottlenecks. For instance, MambaHSI [27] introduces spatial and spectral Mamba-based modeling modules. Wavelet decomposition-based spectral–spatial Mamba [28] further incorporates wavelet decomposition into the Mamba framework. These methods verify Mamba can effectively model long-range spectral–spatial dependencies in single-source HSI classification with linear complexity. However, they do not explicitly address heterogeneous interaction between HSI and auxiliary modalities such as LiDAR. To this end, a hybrid Mamba-Transformer framework [29] has been proposed for joint classification of HSI and LiDAR. However, its multiscale structure may still lead to high memory consumption.

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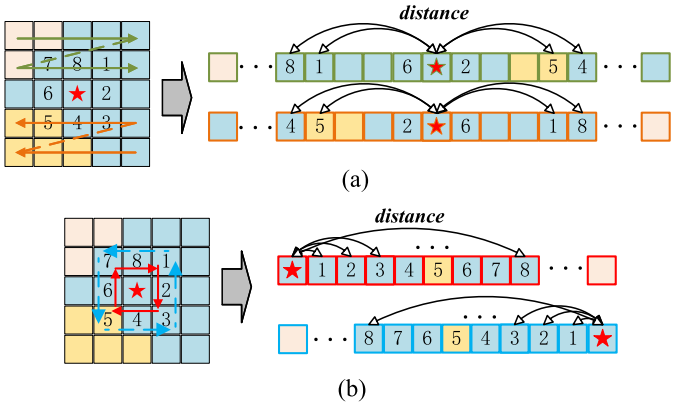


Fig. 2. Selective scanning schemes of Mamba. Different colors denote different categories, and the pentagram indicates the position of the central pixel. (a) Row- or column-major order scanning and its reverse scanning. (b) Sequential or reverse rotation scanning starting from the central pixel.

Furthermore, a frequency-aware hierarchical Mamba [30] was introduced. Although this approach alleviates the disruption of local spatial structures caused by the image flattening process in Mamba through frequency-domain enhancement, the frequency transformation itself introduces additional computational overhead. Despite the breakthroughs achieved by the Mamba architecture in single-source tasks, its application in multimodal tasks remains underexplored. Notably, Mamba was introduced for the first time into the joint classification of HSI and LiDAR data [31]. It was demonstrated that not only can computational complexity be reduced, but also classification performance superior to traditional Transformer methods can be achieved. Additionally, He et al. [32] further proposed a slice Mamba based on a slice-scanning mechanism. By extracting and fusing spatial–spectral features through horizontal and vertical slicing, promising performance and robustness have been demonstrated across multiple datasets. This design is effective for modeling axis-aligned spatial–spectral structures. However, the multipath scanning mechanism increases both computational and structural complexity. To enhance efficiency, a spatial–spectral cross-Mamba network (S^2CrossMamba) [33] was designed. Although effective, this design does not explicitly investigate how different scan orders affect the preservation of the central-pixel neighborhood structure in patch-based HSI-LiDAR classification.

The selective scanning mechanism serves as a core component enabling efficient sequence modeling within the Mamba framework, and the design of its scanning path directly influences the model's capacity to perceive and represent the 2-D spatial structure of an image. Among existing studies, spatial row- or column-wise scanning is commonly adopted as a fundamental strategy [29], [34], [35], [36], [37]. For instance, a 3-D spectral–spatial Mamba (3DSS-Mamba) [38] framework was proposed, where five scanning paths were constructed to investigate the impact of dimensional priority on modeling. However, when the base Mamba model is directly applied to image classification, a fundamental mechanistic limitation becomes apparent: the efficient selective scanning is inherently designed for 1-D sequences. In existing methods, 2-D image pixels are typically flattened into a 1-D sequence using fixed row-major or column-major strategies. As illustrated in Fig. 2(a), this content-agnostic forced linearization process inevitably disrupts the inherent 2-D adjacency relationships among pixels. To mitigate this issue, some studies have explored the expansion of scanning directions. Unidirectional, Sheng et al. [39] investigated bidirectional, four-directional oriented, and full-range scanning, where scanning sequences from different directions were progressively incorporated to enhance the capture of global spatial information. Under raster scanning orders, the serialization of image patches still tends to cross the boundaries of different semantic regions, which disrupts local spatial continuity and introduces memory decay effects in long-range dependency modeling. To address this, a superpixel prior information and an importance-aware reordering strategy were proposed [40], which effectively alleviate the forgetting effect.

Nevertheless, these methods still fail to fully adapt to pixel-level classification tasks. In such tasks, training uses image patches, but the goal is to label the central pixel based on its anisotropic spatial neighborhood. Fixed scanning sequences cannot adaptively focus on this neighborhood, leading to incomplete modeling of discriminative contextual information and a persistent gap between traditional Mamba scanning and the requirements of joint classification.

To address these challenges, this article proposes a lightweight multimodal Mamba (LWM-Mamba) network. Unlike existing cross-Mamba-based fusion frameworks that mainly rely on dedicated cross-modal branches for feature interaction. Specifically, we introduce rotation and reverse-rotation scanning paths centered on the target pixel, as shown in Fig. 2(b). These scanning strategies systematically cover the surrounding neighborhood in a radial manner, enabling more natural and comprehensive modeling of spatial context. Combined with conventional row–column scanning, the proposed multipath mechanism aggregates information from multiple geometric perspectives, leading to more robust spatial dependency modeling. Meanwhile, the lightweight design of LWM-Mamba is achieved through carefully devised shallow embedding and frequency-domain enhancement, ensuring effective extraction and fusion of multimodal features with low parameter counts and computational overhead.

The main contributions of this article can be summarized as follows.

1) A multipath scanning mechanism is proposed, in which conventional scanning is integrated with rotation scanning originating from the central pixel to achieve multidirectional coverage, thereby significantly enhancing spatial context modeling capability.
2) A multimodal adaptive Mamba (MA-Mamba) module is designed, in which the state space matrices of HSI and LiDAR are fused through interaction and dynamic weighting mechanisms to achieve deeper cross-modal collaborative modeling.
3) A frequency enhancement module is proposed to selectively enhance features in the frequency domain, which, combined with spatial residual compensation, improves both local details and global structural representation.

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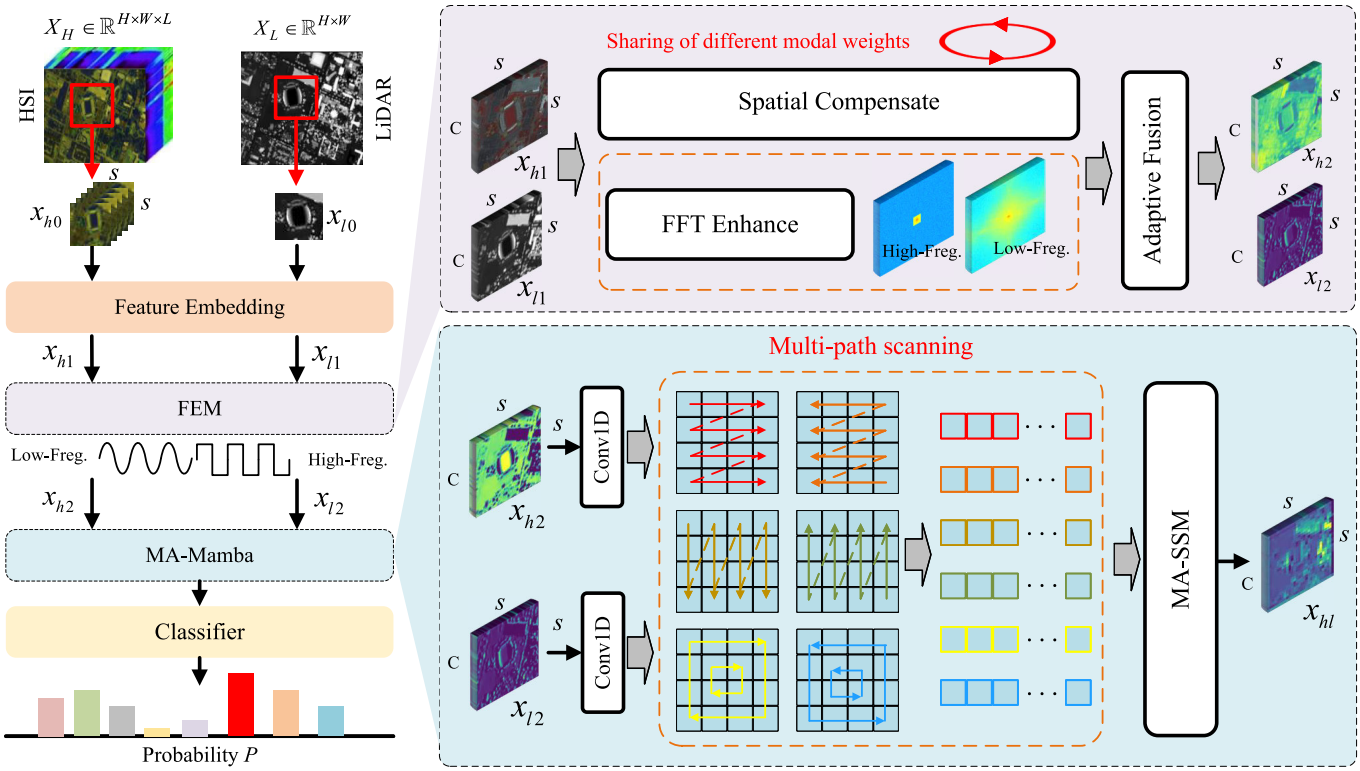


Fig. 3. Overall framework of LWM-Mamba. The network consists of three main modules: feature embedding, FEM, and MA-Mamba.

4) A globally lightweight multimodal framework is constructed, achieving superior classification performance while significantly reducing model parameters and computational complexity.

The remainder of this article is organized as follows. Section II provides a detailed introduction to the overall framework and main modules of LWM-Mamba. Section III presents a series of experimental results and corresponding analyses. Finally, Section IV concludes this article and discusses future work.

II. Methodology

LWM-Mamba is specifically designed for the joint classification of HSI and LiDAR data, aiming to achieve effective multimodal feature fusion and accurate classification. As illustrated in Fig. 3, the network first employs a feature embedding module to extract shallow features from each modality. Subsequently, frequency enhancement and spatial compensation are applied to improve feature representation. The enhanced multimodal features are then fed into the core MA-Mamba module. This module leverages a multipath scanning strategy to capture long-range spatial–spectral dependencies and utilizes 1-D convolution together with the multimodal adaptive SSM (MA-SSM) to achieve deep cross-modal interaction and fusion. Finally, the fused high-level features are passed to a classifier to generate classification probabilities.

A. Feature Embedding

To enable efficient collaborative processing of HSI and LiDAR data while addressing the computational bottleneck in multimodal fusion, a lightweight feature embedding module based on 3-D and 2-D pointwise convolutions (PWConv) is first constructed. Specifically, given the HSI and LiDAR inputs denoted as $X_H \in \mathbb{R}^{H\times W\times L}$ and $X_L \in \mathbb{R}^{H\times W\times 1}$, they are partitioned pixel-wise into patches of spatial size $s \times s$, yielding n sample pairs, where $n = H \times W$. Each sample pair is represented as $x_{h0} \in \mathbb{R}^{s\times s\times L}$ and $x_{l0} \in \mathbb{R}^{s\times s\times 1}$.

For the HSI data, 3-D-PWConv is adopted, which greatly reduces computational complexity without compromising spatial structural information. The process can be formulated as

$$x_{h1} = f_{2d}\left(f_{3d}\left(x_{h0}\right)\right) \tag{1}$$

where $f(\cdot) = \sigma(BN(\cdot))$, with $\sigma(\cdot)$ denoting the activation function and $BN(\cdot)$ representing batch normalization. As indicated by the formulation, the initial HSI patch x_{h0} is passed through two PWConv layers, by which feature dimensionality alignment between modalities is achieved.

For LiDAR elevation data, 2-D PWConv is used for feature transformation, which performs no aggregation operations in the spatial dimensions but linearly combines and remaps the channel dimensions of the input features. The process is expressed as

$$x_{l1} = f_{2d}\left(x_{l0}\right) \tag{2}$$

where x_{l1} denotes the output after 2-D-PWConv.

The use of PWConv as the core operator in the feature embedding stage is key to achieving an overall lightweight network design. Through this module, discriminative spectral–spatial joint features are extracted from HSI, and structured elevation–texture features are obtained from LiDAR.

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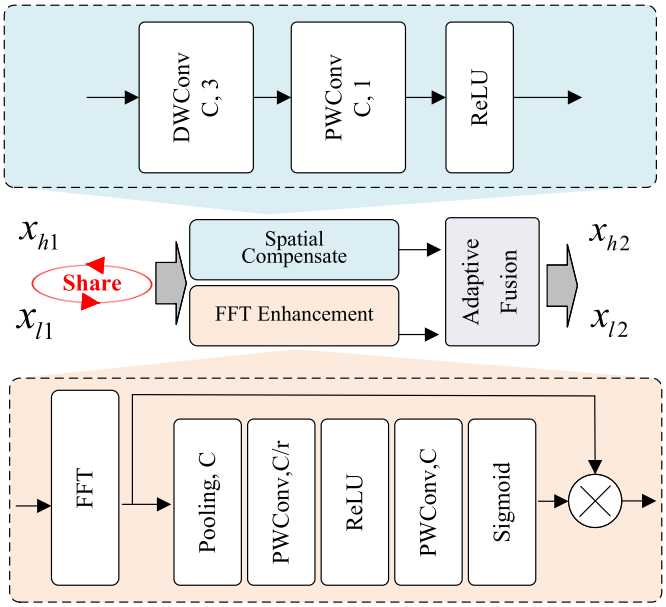


Fig. 4. Structure of FEM.

B. FEM

After obtaining lightweight initial embedded features, a frequency enhancement module (FEM) is designed to further improve feature discriminability. Without significantly increasing model complexity, FEM effectively captures both multiscale details and global structural information, which correspond to high-frequency and low-frequency components, respectively, while introducing a spatial compensation mechanism to preserve feature integrity.

FEM processes multimodal features in parallel with shared parameters. For an input feature from any modality, denoted as $x \in \mathbb{R}^{s\times s\times C}$, the processing pipeline consists of three steps: frequency enhancement, spatial compensation, and adaptive fusion, as illustrated in Fig. 4.

1) Frequency Enhancement: To enhance different frequency components, the input features from each modality are first projected into the frequency domain via the Fast Fourier Transform (FFT). A learnable frequency-domain modulation mechanism is then applied to adaptively emphasize informative frequency components.

Specifically, given the frequency representation $\mathcal{F}(x)$, global average pooling is first applied, followed by a 2-D PWConv-based channel compression to generate a frequency modulation vector. Taking the HSI feature x_{h2} as an example, the process can be formulated as

$$x'_{h2} = \mathcal{F}(x_{h1}) \odot \text{Sigmoid}\left(W_2\delta\left(W_1\text{GAP}(x_{h1})\right)\right) \tag{3}$$

where $\odot$ denotes the element-wise multiplication, $GAP(\cdot)$ denotes the global average pooling, and $\delta(\cdot)$ is the ReLU activation function. The learnable matrices $W_1 \in \mathbb{R}^{C/r\times C}$ and $W_2 \in \mathbb{R}^{C\times C/r}$ correspond to channel reduction and expansion, respectively, where r is the reduction ratio. This mechanism enables adaptive enhancement of either high-frequency or low-frequency components.

2) Spatial Compensation: Although frequency enhancement effectively captures global features, the FFT operation may lead to the loss of local spatial continuity and fine details. To address this issue, a spatial compensation branch is introduced. This branch employs depthwise convolution (DWConv) and PWConv to extract local spatial features efficiently. The process is defined as

$$x''_{h2} = PWConv\left(DWConv_{3\times3}(x_{h1})\right) \tag{4}$$

where $DWConv_{3\times3}(\cdot)$ denotes the DWConv with a kernel size of 3×3, and PWConv performs the channel-wise information fusion. This branch focuses on preserving and enhancing the original local spatial structures and fine-grained details, thereby complementing the frequency-domain representation.

3) Adaptive Fusion: To dynamically integrate the frequency-enhanced features and the spatially compensated features, an adaptive fusion mechanism is introduced. The fusion process is expressed as

$$x_{h2} = \lambda \odot x'_{h2} + (1-\lambda) \odot x''_{h2} \tag{5}$$

where λ denotes a learnable parameter.

This fusion strategy allows the network to adaptively assign weights to each spatial location and channel based on the input content, thereby achieving an optimal balance between global frequency information and local spatial details.

Similarly, LiDAR features can be processed using (3)–(5) to obtain enhanced representations x_{l2}.

C. MA-Mamba

The fusion of deep features from HSI and LiDAR data is a pivotal challenge in multisource joint classification. To address the limitations of existing Mamba-based methods in scanning strategies and cross-modal interaction, the MA-Mamba module is proposed. By introducing a multipath scanning strategy, this module preserves spatial structural integrity and achieves deep interaction within the state space via a learnable cross-modal parameter fusion mechanism. While maintaining linear computational complexity, MA-Mamba significantly enhances the model's capacity for collaborative modeling of contexts and multimodal images. Unlike S^2CrossMamba, which performs cross-modal interaction through a dedicated cross-SSM/IBCM structure, MA-Mamba couples HSI and LiDAR by adaptively fusing their state-space parameters during selective scanning. This design enables modality interaction within the SSM parameterization while maintaining a lightweight architecture.

As illustrated in Fig. 5, the MA-Mamba module comprises two primary stages: multipath scanning and MA-SSM.

1) Stage of Multipath Scanning: As shown in the structure, the multimodal data are first processed via 2-D convolutions, followed by 2-D scanning using paths $\mathcal{P} = \{P_1, P_2, \ldots, P_K\}$. Diverging from traditional Mamba scanning patterns, innovative rotation and inverse rotation scanning paths originate from the central pixel (c_x, c_y). For the former, starting from the center pixel, the neighborhood pixels are traversed layer by layer from the inside out in a clockwise direction, generating a progressive perception sequence from local details to global

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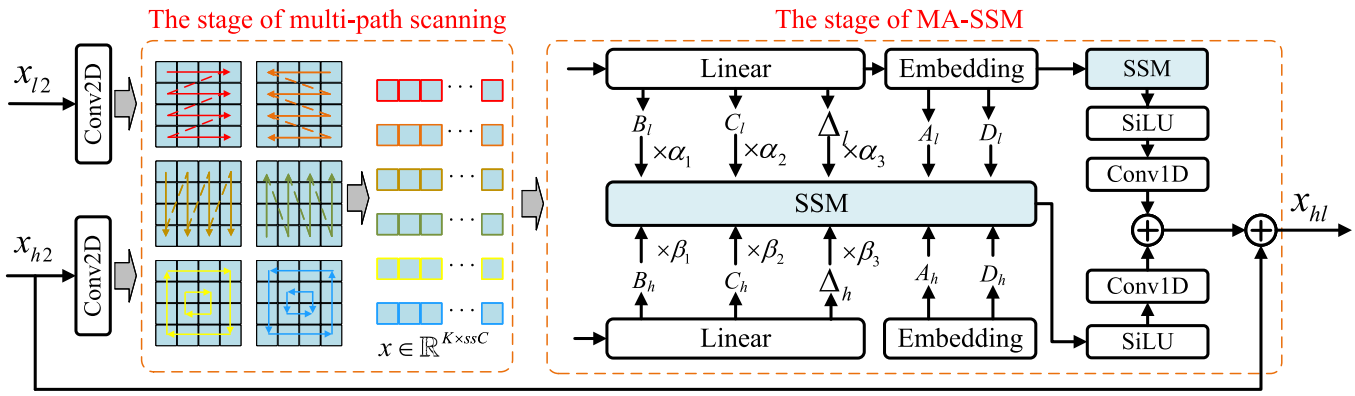


Fig. 5. Structure of MA-Mamba.

structure. For the latter, the traversal proceeds counterclockwise from the outer edges inward to the center pixel, enhancing the constraint of global structure on the central semantics. From the perspective of receptive field consistency, conventional row- or column-major scanning treats the entire image patch as a flat sequence, ignoring the fact that the classification of the central pixel depends on a locally anisotropic neighborhood. In remote sensing imagery, the contextual information around a central pixel typically exhibits a radial distribution pattern, where adjacent pixels contribute more to the semantic label than farther ones, and the importance varies with direction. Consequently, the proposed rotation and reverse-rotation paths better preserve the 2-D spatial structure and yield more discriminative state representations for pixel-level classification.

For any spatial position (i, j), its sequence index $pos_k(i, j)$ in the kth path is determined by the specific scanning function. The serialization of distinct scanning paths K yields a set of feature representations $\{S_1, S_2, \ldots, S_K\}$, where $S_k \in \mathbb{R}^{ssC}$. Consequently, the input features x_{h2} and x_{l2} are transformed into sequences K, denoted as $\{S_k^H\}_{k=1}^K$ and $\{S_k^L\}_{k=1}^K$.

2) Stage of MA-SSM: This stage involves the collaborative processing of multimodal sequence pairs from the same path. A traditional SSM for discrete sequences is formulated as a linear time-invariant system. The process can be formulated as

$$h_t = \overline{A}h_{t-1} + \overline{B}s_t, \quad y_t = Ch_t \tag{6}$$

where s_t is the input at time t, h_t is the hidden state, and $\overline{A}, \overline{B}, \overline{C}$ are system parameters. To achieve deep cross-modal fusion, this is extended into an adaptive state space fusion mechanism. Instead of calculating states independently, interaction during state updates is implemented via a dynamic fusion gate. For HSI, the parameters are generated through linear projections of its own sequence. The process can be formulated as

$$\left(\Delta_k^H, A_k^H, B_k^H, C_k^H\right) = \text{Linear}_{\theta H}\left(S_K^H\right). \tag{7}$$

Similarly, the LiDAR modality generates its parameters as

$$\left(\Delta_k^L, A_k^L, B_k^L, C_k^L\right) = \text{Linear}_{\theta L}\left(S_K^L\right). \tag{8}$$

Here, Δ_k is a time step parameter that depends on the input and is used to discretize the continuous system parameters A, B into $\overline{A}_k, \overline{B}_k$. The discretization process follows the zero-order hold. The process is expressed as

$$\begin{cases} \overline{A}_k^H = \exp\left(\Delta_k^H \cdot A_k^H\right) \\ \overline{B}_k^H = \exp\left(\Delta_k^H \cdot A_k^H\right)^{-1}\left(\exp\left(\Delta_k^H \cdot A_k^H\right) - I\right) \cdot \Delta_k^H \cdot B_k^H \\ \overline{A}_k^L = \exp\left(\Delta_k^L \cdot A_k^L\right) \\ \overline{B}_k^L = \exp\left(\Delta_k^L \cdot A_k^L\right)^{-1}\left(\exp\left(\Delta_k^L \cdot A_k^L\right) - I\right) \cdot \Delta_k^L \cdot B_k^L. \end{cases} \tag{9}$$

To fully leverage the precise spatial structural information provided by LiDAR for enhancing the spectral feature modeling of HSI, learnable parameters $\alpha = \{\alpha_1, \alpha_2, \alpha_3\}$ and $\beta = \{\beta_1, \beta_2, \beta_3\}$ are employed to adaptively fuse the LiDAR system parameters $\{\overline{A}_k^L, \overline{B}_k^L, \Delta_k^L\}$ with the HSI parameters. The process is expressed as

$$\begin{cases} \tilde{A}_k^H = \alpha_1 \overline{A}_k^H + \beta_1 \overline{A}_k^L \\ \tilde{B}_k^H = \alpha_2 \overline{B}_k^H + \beta_2 \overline{B}_k^L \\ \tilde{\Delta}_k^H = \alpha_3 \Delta_k^H + \beta_3 \Delta_k^L. \end{cases} \tag{10}$$

Unlike feature-level concatenation, which directly stacks multimodal features and is sensitive to noise and modality discrepancies, our parameter-level fusion integrates the system parameters $(\overline{A}, \overline{B}, \Delta)$ of each modality. These parameters encode the global sequential dynamics of the respective modality, acting as a robust low-pass representation that suppresses local noise and adaptively balances cross-modal contributions through learnable weights α, β. The fused parameters are then used to compute the initial updated hidden state $\tilde{h}_t^{H,fused}$ for the HSI sequence. For the LiDAR sequence $\tilde{h}_t^L$, the state update primarily utilizes its own parameters $\{\overline{A}_k^L, \overline{B}_k^L, \overline{\Delta}_k^L\}$. The process is expressed as

$$\begin{cases} \tilde{h}_t^{H,fused} = \tilde{A}_k^H \tilde{h}_{t-1}^{H,fused} + \tilde{B}_k^H S_t^H, & \tilde{y}_t^H = C_t^H \tilde{h}_t^{H,fused} \\ \tilde{h}_t^L = \overline{A}_k^L \tilde{h}_{t-1}^L + \overline{B}_k^L S_t^L, & \tilde{y}_t^L = C_t^L \tilde{h}_t^L. \end{cases} \tag{11}$$

Finally, to incorporate spectral context, a gating mechanism $G_t(x_{h2})$ is designed for the final fusion. The process is expressed as

$$\tilde{y}_t^{hl} = G_t + \left(Conv1D\left(SiLU\left(\tilde{y}_t^L\right)\right) + Conv1D\left(SiLU\left(\tilde{y}_t^H\right)\right)\right). \tag{12}$$

Here, Conv1D and SiLU are used for feature transformation and nonlinear enhancement. For each path P_k, after the above

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MA-SSM processing, the sequence representation after multimodal fusion is obtained. It is then restored to a 2-D feature x_k^{hl} through the inverse process of the corresponding scanning path. To aggregate the diverse spatial context views provided by K different paths, the outputs of all paths are further fused to obtain the final output $x^{hl} = \{x_1^{hl}, x_2^{hl}, \ldots, x_K^{hl}\}$.

Algorithm 1 Training and Testing of LWM-Mamba

Input:

Labeled sample sets X_h and X_l for training, test sample sets X_{h_t} and X_{h_t}, total number of training iterations T, and corresponding labels Y. Mini-batch size $B = 64$, learning rate $\eta = 10^{-3}$, optimizer: Adam.

Procedure:

Initialize the network parameters using a Gaussian distribution.
for $iter = 0$ **to** T **do**
  Randomly sample a mini-batch of pairs (x_{h0}, x_{l0}).
  Perform Feature embedding via Eqs. (1)–(2) to obtain x_{h1}, x_{l1}.
  Apply FEM via Eqs. (3)–(5) to obtain x_{h2} and x_{l2}.
  Execute Multipath scanning to generate serialized feature sequences.
  Update the MA-SSM parameters.
  Compute the classification loss $\mathcal{L}_{cls} = CrossEntropy(f(x_{hi}), Y_{hi})$ using the forward propagation output.
  Calculate the gradient $\nabla_\theta \mathcal{L}_{cls}$ via backpropagation.
  Update all network parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{cls}$, where η is the learning rate.
end for
Input test pairs (X_{h_t}, X_{l_t}) into the trained LWM-Mamba network.
Perform forward propagation to predict categories for each sample.

Output:

Classification performance metrics: OA, AA, K.

3) Implementation Details: The comprehensive training and testing workflow of LWM-Mamba is summarized in Algorithm 1. Input HSI and LiDAR data are processed into pixel-wise patches to form the labeled training set $\{X_h, X_l, Y\}$ and the test set $\{X_{h_t}, X_{l_t}\}$. All learnable parameters are randomly initialized using a Gaussian distribution. The optimization objective is to minimize the cross-entropy classification loss using the Adam optimizer over a total of T iterations.

In the training phase, mini-batches of sample pairs (x_{h0}, x_{l0}) and their corresponding labels y_0 are randomly sampled. After forward propagation through the feature embedding, frequency enhancement, and MA-Mamba modules, the network computes the gradient $\nabla_\theta \mathcal{L}_{cls}$ via backpropagation and updates the parameters using $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{cls}$, where η is the learning rate. For testing, the fixed parameters are applied to the test set to predict the categories, and performance is evaluated using OA, average accuracy (AA), and the Kappa coefficient.

III. Experimental Results and Analysis

A. Dataset Descriptions

Three widely used benchmark datasets, namely Houston, MUUFL, and Trento, are adopted for experimental evaluation. Detailed descriptions of these datasets are provided as follows.

1) *Houston Dataset:* This dataset was collected over the campus of the University of Houston and its surrounding urban areas using the compact airborne spectrographic imager (CASI). The spatial size of the dataset is 345 × 1905 pixels, with a spatial resolution of 2.5 m. The HSI data contain 144 spectral bands, covering a wavelength range from 0.38 to 1.35 μm. In total, 15 029 labeled samples are provided, which are categorized into 15 land-cover classes.
2) *MUUFL Dataset:* This dataset was acquired over the University of Southern Mississippi, Gulfport campus, located in Long Beach, Mississippi, USA. The HSI data have a spatial size of 325 × 220 pixels with 64 spectral bands. The LiDAR data were collected using an ALTM sensor with a wavelength of 1064 nm. The dataset contains 53 687 labeled samples belonging to 11 land-cover classes.
3) *Trento Dataset:* This dataset was collected in a rural area in Trento, southern Italy, and consists of HSI and LiDAR data. The spatial size is 166 × 600 pixels with a spatial resolution of 1 m. The HSI data cover a wavelength range from 0.42 to 0.99 μm with 63 spectral bands. The LiDAR data were acquired using a 3100EA sensor. A total of 30 214 labeled samples are included, corresponding to six land-cover classes.

The detailed class information for the three datasets is summarized in Table I.

B. Experimental Settings

All experiments are implemented using the PyTorch framework and conducted on a server equipped with a 2.1 GHz Intel Xeon Silver 8352V CPU and an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory. The proposed LWM-Mamba model is optimized using the Adam optimizer, with the number of training epochs set to 500. The input spatial patch size is fixed at 11 × 11. To objectively evaluate the performance of different methods, three widely used evaluation metrics are adopted: OA, AA, and the Kappa coefficient.

C. Sensitivity Analysis

1) Adaptive Fusion Coefficient Analysis: To analyze the adaptive fusion behavior, we tracked the learned coefficients α and β, which correspond to the HSI and LiDAR branches, respectively. Both coefficients were initialized as $(0.5, 0.5)$. As shown in Fig. 6, their trajectories evolve differently across datasets, indicating that the fusion module learns dataset-specific modality weights rather than using fixed fusion coefficients. These results demonstrate that LWM-Mamba can adaptively adjust the contributions of HSI and LiDAR according to dataset characteristics, improving the interpretability of the proposed fusion module.

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TABLE I
DETAILED INFORMATION ON THREE DATASETS

No.	Houston			MUUFL			Trento		
	Class name	Train	Test	Class name	Train	Test	Class name	Train	Test
C1	Healthy grass	20	1,231	Trees	20	23,226	Apple	20	4,014
C2	Stressed grass	20	1,234	Mostly Grass	20	4,250	Buildings	20	2,883
C3	Synthetic	20	677	Mixed Ground	20	6,862	Ground	20	459
C4	Trees	20	1,224	Dirt and Sand	20	1,806	Woods	20	9,103
C5	Soil	20	1,222	Road	20	6,667	Vineyard	20	10,481
C6	Water	20	305	Water	20	446	Roads	20	3,154
C7	Residential	20	1,248	Buildings Shadow	20	2,213			
C8	Commercial	20	1,224	Buildings	20	6,220			
C9	Road	20	1,232	Sidewalk	20	1,365			
C10	Highway	20	1,207	Yellow Curb	20	163			
C11	Railway	20	1,215	Cloth Panels	20	249			
C12	Parking Lot 1	20	1,213						
C13	Parking Lot 2	20	449						
C14	Tennis Court	20	408						
C15	Running	20	640						
-	Total	300	14,729	Total	220	53,467	Total	120	30,094

TABLE II
IMPACT OF DIFFERENT MODULES ON CLASSIFICATION PERFORMANCE (OA%)

Feature embedding	FEM	MA-Mamba	Houston	MUUFL	Trento
√			93.10	75.52	97.32
√	√		93.69 (+0.59)	79.96 (+4.44)	98.74 (+1.42)
√	√	√	95.68 (+1.99)	86.43 (+6.47)	99.46 (+1.06)

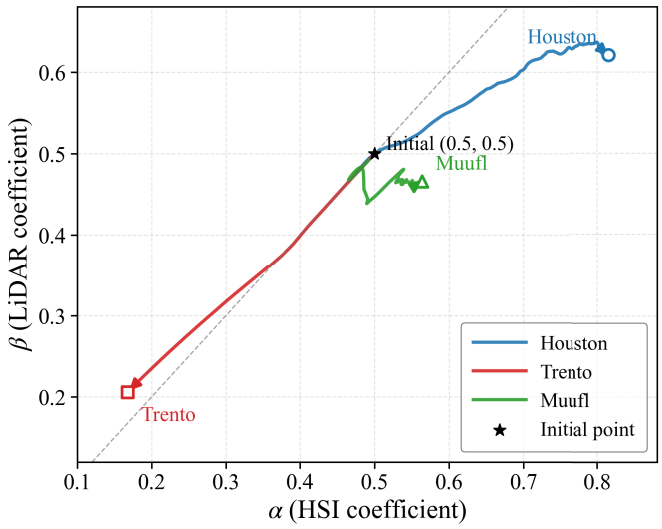


Fig. 6. Trajectories of the learned fusion coefficients α and β on Houston, Trento, and MUUFL datasets. The initial point is (0.5, 0.5), where α and β denote the learned contributions of the HSI and LiDAR branches, respectively.

2) Ablation Study: The proposed LWM-Mamba consists of three main components: the feature embedding module, FEM, and the MA-Mamba module. To evaluate the effectiveness of each component, ablation experiments are conducted on the Houston, MUUFL, and Trento datasets, with the results summarized in Table II. As observed, when only the feature embedding module is employed, the model achieves OA values of 93.10%, 75.52%, and 97.32% on the Houston, MUUFL, and Trento datasets, respectively. After incorporating the FEM module, performance improvements are observed across all three datasets. This indicates that FEM effectively enhances feature representation through its frequency-domain enhancement and spatial compensation mechanisms. Notably, an improvement of 4.44% is achieved on the MUUFL dataset, demonstrating its strong adaptability in complex scenarios. Furthermore, when the MA-Mamba module is introduced, the performance is further improved across all datasets. Compared with the combination of feature embedding and FEM, a more significant improvement is observed on the MUUFL dataset, highlighting the effectiveness of MA-Mamba in multimodal feature fusion and contextual dependency modeling. This improvement also verifies the advantages of the proposed multipath scanning mechanism and MA-SSM in joint classification tasks. Overall, the inclusion of each module consistently contributes to performance gains, demonstrating good consistency and robustness across different datasets.

3) Hyperparameter Analysis: To evaluate the sensitivity of the proposed LWM-Mamba to key hyperparameters, classification performance in terms of OA is examined under different combinations of learning rates and batch sizes on the Houston, MUUFL, and Trento datasets. The results are illustrated in Fig. 7. In the experiments, the learning rate is selected from $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$, while the batch size is chosen from {16, 32, 64, 128}. As observed in Fig. 7, the surfaces for all three datasets exhibit relatively flat trends, indicating that the model performance remains stable within the tested hyperparameter space. This stability can be mainly attributed to the effectiveness of the FEM and the

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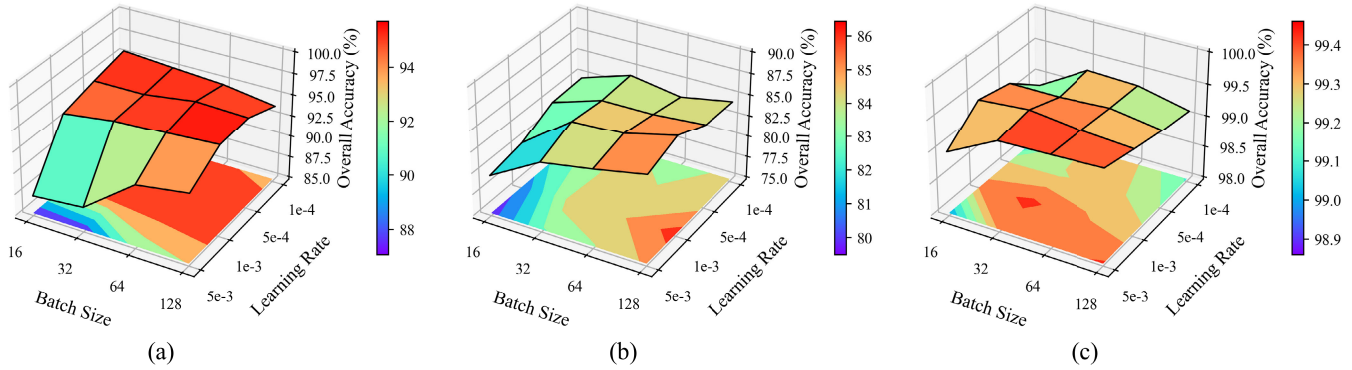


Fig. 7. Impact of different learning rates and batch sizes on performance. (a) Houston dataset. (b) MUUFL dataset. (c) Trento dataset.

TABLE III
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON HOUSTON DATASET. (BEST RESULTS SHOWN IN BOLD)

Class	HSI+LiDAR								Our
	HCT 2022	MFT 2023	GAMF 2024	Cross-HL 2024	HMAT 2025	MCFNet 2025	S^2CrossMamba 2025	LSLFormer 2026	LWM-Mamba
C1	92.93	96.91	**99.59**	97.97	85.05	97.40	90.75	93.03	95.13
C2	98.62	94.97	97.33	89.87	87.03	96.11	98.68	85.33	**99.92**
C3	99.85	**100**	99.85	98.82	98.81	98.37	99.69	89.37	**100**
C4	99.34	91.33	93.87	99.75	**100**	97.87	96.01	90.87	99.02
C5	99.67	97.70	99.67	**100**	**100**	99.83	99.75	86.53	99.26
C6	98.68	**100**	92.13	94.09	**100**	95.40	**100**	80.55	98.36
C7	**98.71**	76.84	81.41	91.51	98.07	87.58	39.49	71.10	96.47
C8	76.87	71.32	85.94	77.04	83.82	77.45	**96.84**	73.73	87.58
C9	76.13	82.22	79.38	85.95	84.98	**93.26**	28.05	51.29	83.52
C10	68.68	94.03	74.15	97.10	91.05	94.78	74.22	47.95	**97.43**
C11	94.89	89.95	87.48	81.64	93.41	**100**	88.20	57.42	96.63
C12	46.66	92.82	87.88	92.91	92.08	82.35	44.42	42.46	**93.73**
C13	97.55	98.44	93.76	95.32	95.54	92.87	**100**	32.30	**100**
C14	**100**	99.26	99.26	**100**	99.01	**100**	99.74	78.42	**100**
C15	99.37	91.09	**100**	**100**	**100**	**100**	80.32	92.91	**100**
OA	87.67	90.22	90.21	92.50	92.76	93.51	78.74	71.99	**95.68**
AA	89.87	91.80	91.45	93.47	93.93	94.22	82.41	72.69	**96.44**
K×100	86.69	89.43	89.41	91.90	92.17	92.98	77.23	71.99	**95.33**

multipath scanning mechanism, which enable robust extraction of discriminative features under different training settings. When comparing the three datasets, the highest OA is achieved on the Trento dataset, followed by MUUFL, while relatively lower performance is observed on the Houston dataset. This difference is consistent with the inherent characteristics of the datasets. Specifically, the Trento dataset contains fewer classes with clearer structural patterns, making it easier to classify, whereas the Houston dataset involves more diverse categories and more complex scenes, leading to increased classification difficulty and comparatively lower accuracy. Overall, the proposed LWM-Mamba demonstrates consistently high accuracy and stable performance across a wide range of hyperparameter settings, highlighting its strong robustness and generalization capability.

D. Experimental Comparison

To comprehensively evaluate the performance of the proposed LWM-Mamba, comparisons are conducted with several state-of-the-art methods on three datasets. The compared methods include HCT [16], MFT [21], GAMF [19], Cross-HL [22], HMAT [20], MCFNet [23], S^2CrossMamba [33], and LSLFormer [25].

1) Quantitative Evaluation:

a) Houston dataset: As shown in Table III, the proposed LWM-Mamba achieves an OA of 95.68%, an AA of 96.44%, and a Kappa of 95.33%, all of which outperform the compared methods by a clear margin. The suboptimal method is MCFNet, and the proposed method achieves an improvement of 2.17% in OA over it. In contrast, earlier methods such as HCT and MFT obtain OA values below 91%. Among the 15 land-cover classes, LWM-Mamba achieves the best or comparable performance in seven classes, including C2, C3, and C13–C15. Notably, perfect classification accuracy is achieved for C3 and C13–C15. Although slightly lower accuracy is observed for a few categories such as C8 and C9 compared with individual methods, the overall superiority remains evident. These results demonstrate that the proposed multipath scanning mechanism and adaptive fusion strategy are capable of effectively handling the challenges

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TABLE IV
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON THE MUUFL DATASET. (BEST RESULTS SHOWN IN BOLD)

Class	HSI+LiDAR								Our
	HCT 2022	MFT 2023	GAMF 2024	Cross-HL 2024	HMAT 2025	MCFNet 2025	S^2CrossMamba 2025	LSLFormer 2026	LWM-Mamba
C1	85.11	88.94	89.99	89.39	91.33	90.64	85.68	78.37	**91.64**
C2	72.18	75.97	75.01	70.87	80.07	71.38	**80.35**	73.88	78.45
C3	64.83	59.80	65.06	71.45	53.30	71.32	**74.89**	37.96	70.80
C4	**93.68**	81.00	90.75	90.69	84.71	92.96	93.63	78.84	88.59
C5	72.35	69.11	**88.34**	82.61	83.23	85.94	76.93	80.29	86.17
C6	99.55	99.77	99.77	**100**	99.77	**100**	97.76	91.70	**100**
C7	84.90	85.04	93.04	**95.30**	89.60	84.31	85.90	85.17	90.60
C8	93.24	85.77	94.11	93.29	**96.72**	95.80	88.04	94.63	92.54
C9	53.26	45.71	51.57	60.95	54.28	45.87	46.23	40.29	**61.25**
C10	74.33	71.77	84.66	69.32	74.23	71.11	**79.14**	52.76	64.42
C11	99.59	95.98	95.58	99.59	92.36	90.76	**99.60**	84.33	96.79
OA	80.46	79.87	85.14	84.86	83.96	85.40	82.47	74.35	**86.43**
AA	81.18	78.08	**84.36**	83.95	81.79	82.05	82.56	72.57	83.75
K×100	75.06	74.08	80.79	80.39	79.22	81.06	77.51	67.68	**82.35**

TABLE V
CLASSIFICATION ACCURACY (%) OBTAINED BY DIFFERENT METHODS ON TRENTO DATASET. (BEST RESULTS SHOWN IN BOLD)

Class	HSI+LiDAR								Our
	HCT 2022	MFT 2023	GAMF 2024	Cross-HL 2024	HMAT 2025	MCFNet 2025	S^2CrossMamba 2025	LSLFormer 2026	LWM-Mamba
C1	99.15	97.48	95.26	99.32	99.52	**100**	97.91	97.70	99.75
C2	97.88	98.02	93.09	84.59	**99.68**	98.30	99.10	99.41	99.34
C3	97.82	95.42	**100**	**100**	**100**	97.82	97.81	92.15	98.91
C4	**100**	99.89	**100**	**100**	**100**	**100**	**100**	**100**	**100**
C5	98.73	99.97	98.35	99.89	**100**	**100**	99.99	96.86	**100**
C6	94.83	93.84	93.27	94.99	91.12	94.45	93.72	91.88	**95.97**
OA	98.67	98.72	97.43	97.87	98.98	99.22	98.94	97.57	**99.46**
AA	98.07	97.44	96.67	96.47	98.39	98.43	98.09	96.34	**98.99**
K×100	98.22	98.29	96.58	97.16	98.63	98.96	98.59	96.77	**99.28**

of high class diversity and spectral overlap in the Houston dataset.

b) MUUFL dataset: The comparison results are presented in Table IV. The proposed LWM-Mamba achieves the best performance in terms of OA and Kappa. The AA reaches 83.75%, which is slightly lower than that of GAMF but still superior to other competing methods. Compared with the suboptimal method MCFNet, an improvement of 1.03% in OA is achieved. Across the 11 classes, LWM-Mamba shows strong performance in several categories, such as C1, C6, and C9. In particular, perfect classification is achieved for C6, and significantly higher accuracy is obtained for C9 compared with other methods. Although slightly lower performance is observed for certain classes such as C2 and C3, LWM-Mamba maintains consistently high accuracy across most categories, resulting in the highest overall OA. This indicates that the proposed method exhibits strong robustness when dealing with the complex characteristics of the MUUFL dataset.

c) Trento dataset: The results are listed in Table V. The proposed LWM-Mamba achieves an OA of 99.46%, an AA of 98.99%, and a Kappa of 99.28%, outperforming all compared methods. The suboptimal method is MCFNet, and an improvement of approximately 0.24% in OA is observed. Among the six classes, LWM-Mamba achieves 100% accuracy for C4 and C5. For C1 and C2, accuracies of 99.75% and 99.34% are obtained, respectively, which are close to or exceed the best results of competing methods. Since the Trento dataset contains fewer classes and relatively clear spatial structures, it is generally easier to classify. Nevertheless, further performance improvements are still achieved by the proposed method, demonstrating its strong capability for fine-grained classification even in relatively simple scenarios.

2) Visual Evaluation:

a) Houston dataset: Fig. 8 presents the classification results of all compared methods. This dataset contains 15 land-cover classes with diverse categories, and a large number of regions exhibit irregular shapes and complex boundaries, making accurate classification challenging. From a visual perspective, considerable salt-and-pepper noise can be observed in the results of HCT and MFT. In particular, obvious misclassified patches are present in the Residential and Commercial areas. Although the noise is reduced to some extent by GAMF and Cross-HL, blurred boundaries still remain for several classes. For S^2CrossMamba and LSLFormer, relatively low classification accuracy is achieved, and their results are severely affected by noise. A large number of land-cover pixels are incorrectly labeled, making the true spatial distribution nearly indistinguishable. This observation is consistent with their low OA values of 78.74% and 71.99%, respectively. In contrast, the classification map generated by LWM-Mamba is closest to the ground truth. Homogeneous regions, such as Residential, Commercial, and Road, are represented by

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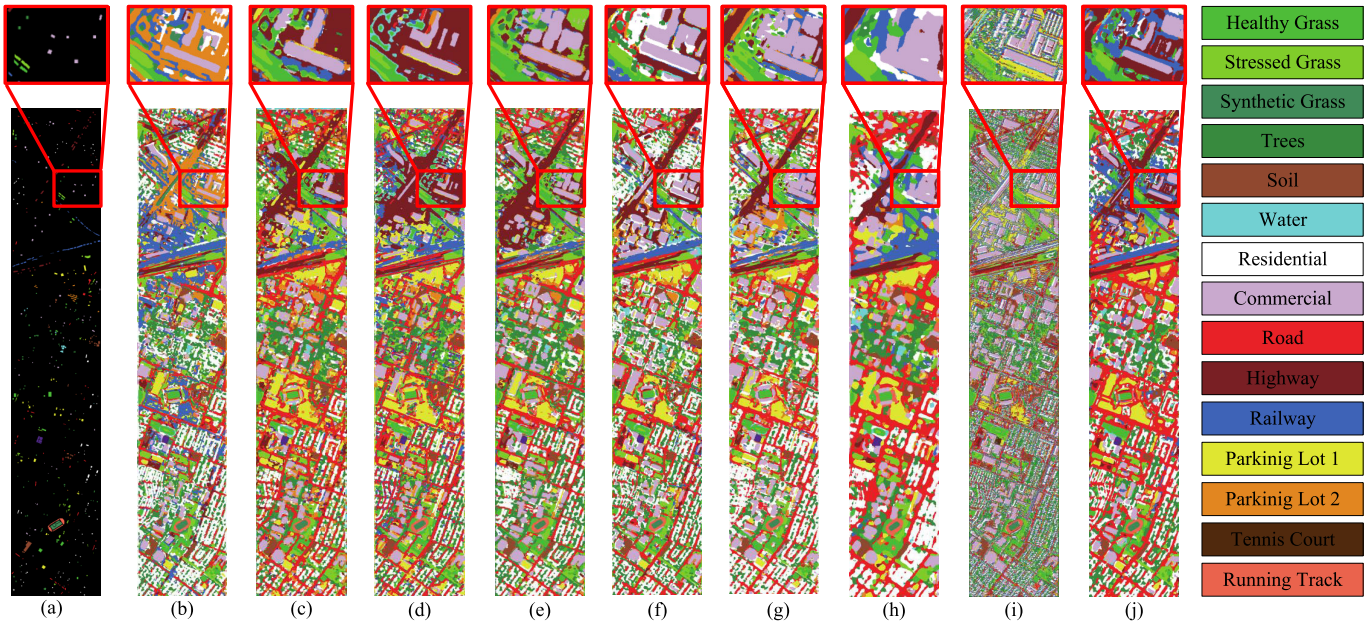


Fig. 8. Classification results obtained by different methods on the Houston dataset. (a) Ground-truth map. (b) HCT (87.67%). (c) MFT (90.22%). (d) GAMF (90.21%). (e) Cross-HL (92.50%). (f) HMAT (92.76%). (g) MCFNet (93.51%). (h) S^2CrossMamba (78.74%). (i) LSLFormer (71.99%). (j) LWM-Mamba (95.68%).

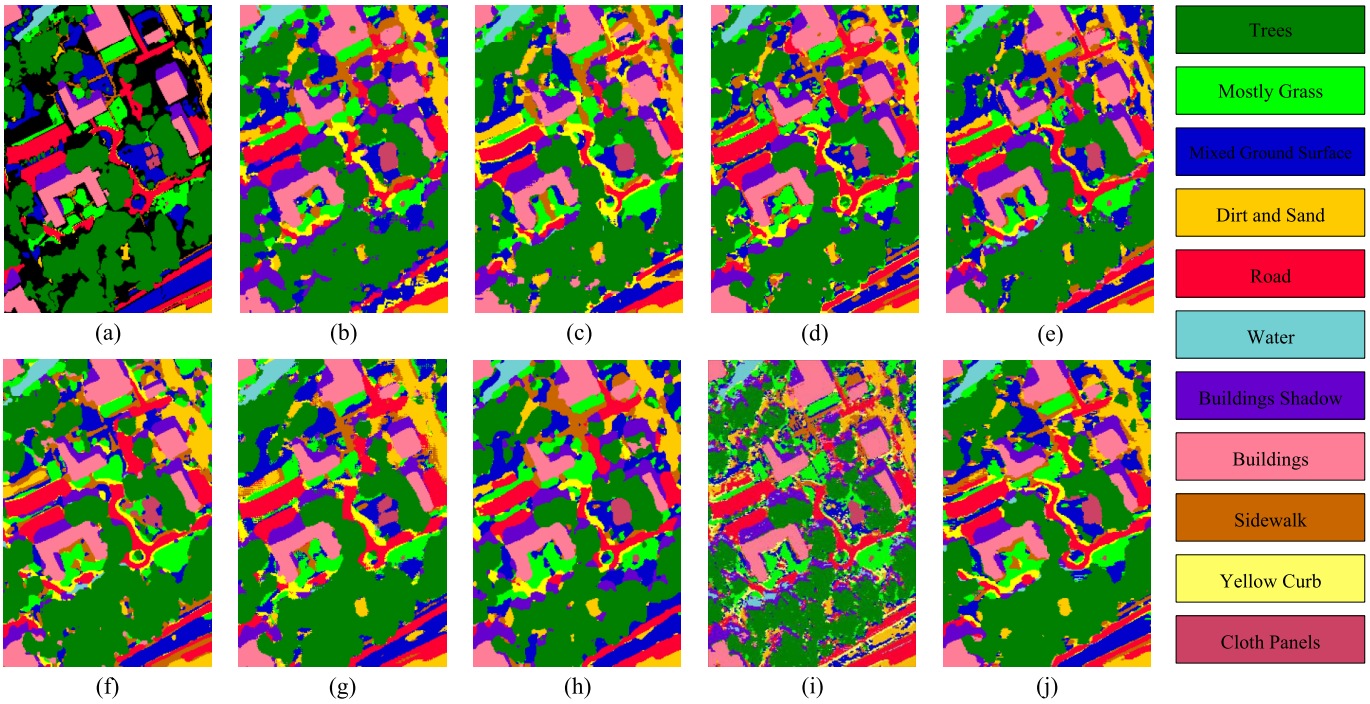


Fig. 9. Classification maps obtained by different methods on the MUUFL dataset. (a) Ground truth. (b) HCT (80.46%). (c) MFT (79.87%). (d) GAMF (85.14%). (e) Cross-HL (84.86%). (f) HMAT (83.96%). (g) MCFNet (85.40%). (h) S^2CrossMamba (82.47%). (i) LSLFormer (74.35%). (j) LWM-Mamba (86.43%).

continuous and clean segments. This improvement can be attributed to the adaptive fusion module, which effectively integrates cross-modal information.

b) MUUFL dataset: Fig. 9 illustrates the classification results. This dataset is characterized by a high proportion of vegetation classes alongside various man-made structures such as "Buildings" and "Sidewalks," resulting in high intraclass spectral similarity. Overall, the classification map produced by LWM-Mamba is the most visually consistent, exhibiting superior intraclass homogeneity and the lowest noise levels compared with other methods. Notably, for classes with scarce and scattered samples such as "Yellow Curb" and "Cloth Panels," LWM-Mamba effectively preserves their structural integrity. This improvement can be attributed to the

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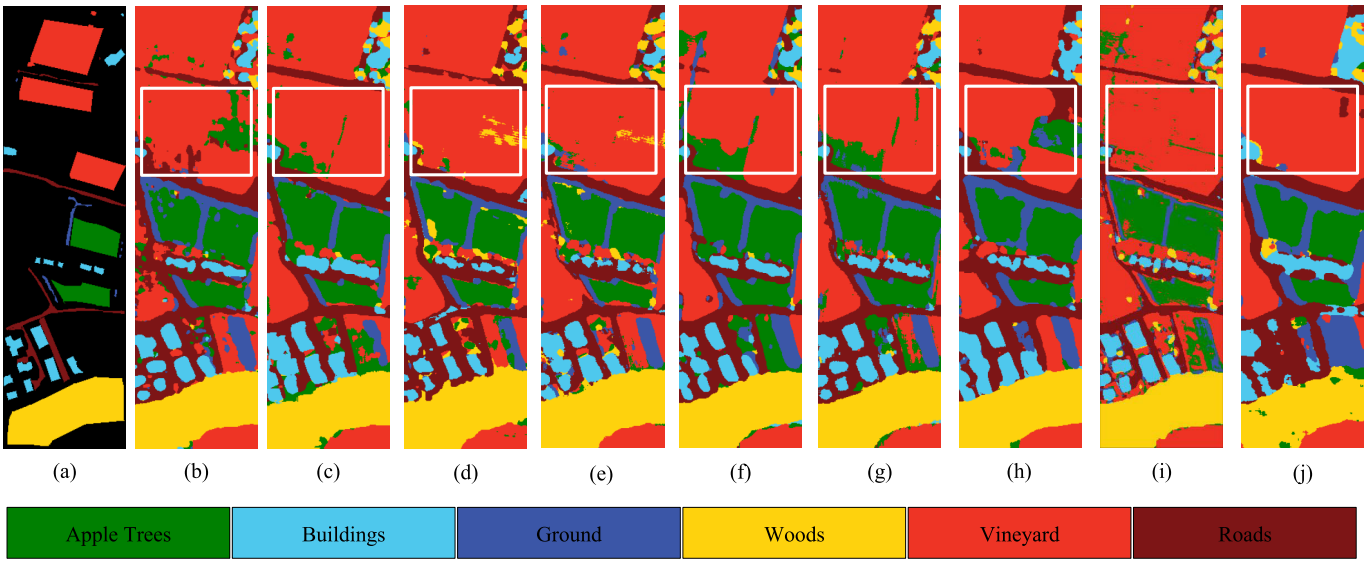


Fig. 10. Classification maps obtained by different methods on the Trento dataset. (a) Ground truth. (b) HCT (98.67%). (c) MFT (98.72%). (d) GAMF (97.43%). (e) Cross-HL (97.87%). (f) HMAT (98.98%). (g) MCFNet (99.22%). (h) S^{2}CrossMamba (98.94%). (i) LSLFormer (97.57%). (j) LWM-Mamba (99.46%).

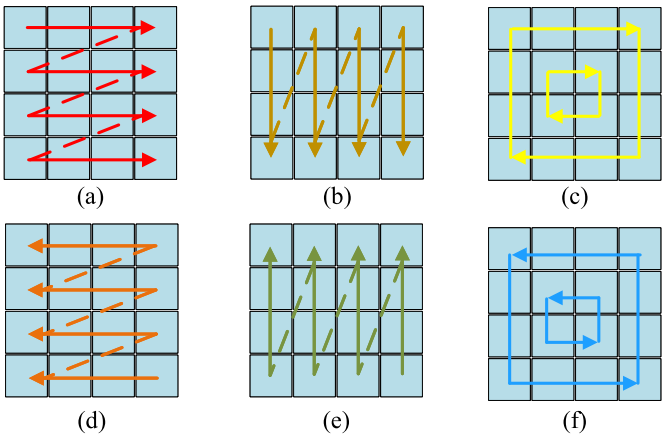


Fig. 11. Multipath scanning mechanism. (a) Row scan. (b) Column scan. (c) Rotation scan. (d) Reverse scan. (e) Reverse column scan. (f) Reverse rotation scan.

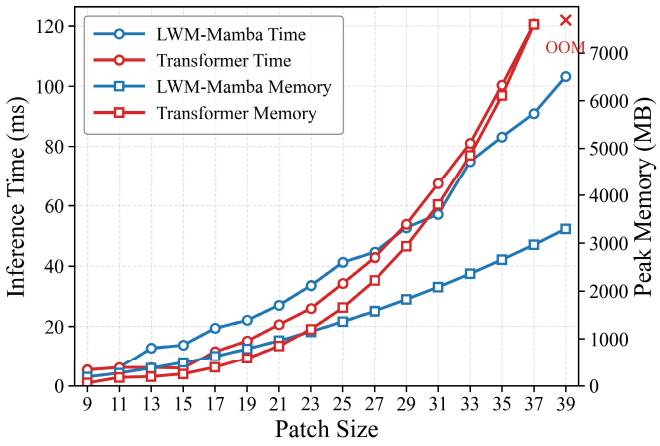


Fig. 12. Forward time and memory complexity comparison between LWM-Mamba and Transformer.

TABLE VI
CONTRIBUTION OF DIFFERENT MULTIPATH SCANNING MECHANISMS (OA%)

Path	Houston	MUUFL	Trento
(a)+(b)	94.19	84.79	98.41
(a)+(b)+(d)+(e)	94.13	85.68	99.13
(a)+(b)+(c)+(d)+(e)+(f)	95.68 (**+1.55**)	86.43 (**+0.75**)	99.46 (**+0.33**)

FEM, which significantly enhances the representation of fine-grained detail features. Furthermore, the boundaries between "Buildings" and "Sidewalks" are clearly defined in the results, with substantially fewer misclassified regions than those of competing methods. This demonstrates the efficacy of the MA-Mamba module in achieving robust cross-modal information integration.

c) Trento dataset: Fig. 10 presents the classification results. Although this dataset is characterized by a limited number of classes and relatively simple ground structures, leading to generally high accuracy across all methods, visual differences remain. Specifically, the classification maps of HCT, MFT, and GAMF exhibit sparse discrete noise within the "Vineyard" regions, accompanied by a slight lack of continuity in the "Roads" areas. While MCFNet and S^{2}CrossMamba yield relatively smooth classification maps, LSLFormer achieves the lowest accuracy, with noticeable fractures in the "Roads" areas and significant misclassifications in the "Ground" areas. In contrast, the classification map of LWM-Mamba shows a high degree of congruence with the ground truth. It is characterized by the absence of visible noise within each class and the presence of sharp, well-defined edges. These results demonstrate that even in simpler scenarios, LWM-Mamba can achieve superior visual quality

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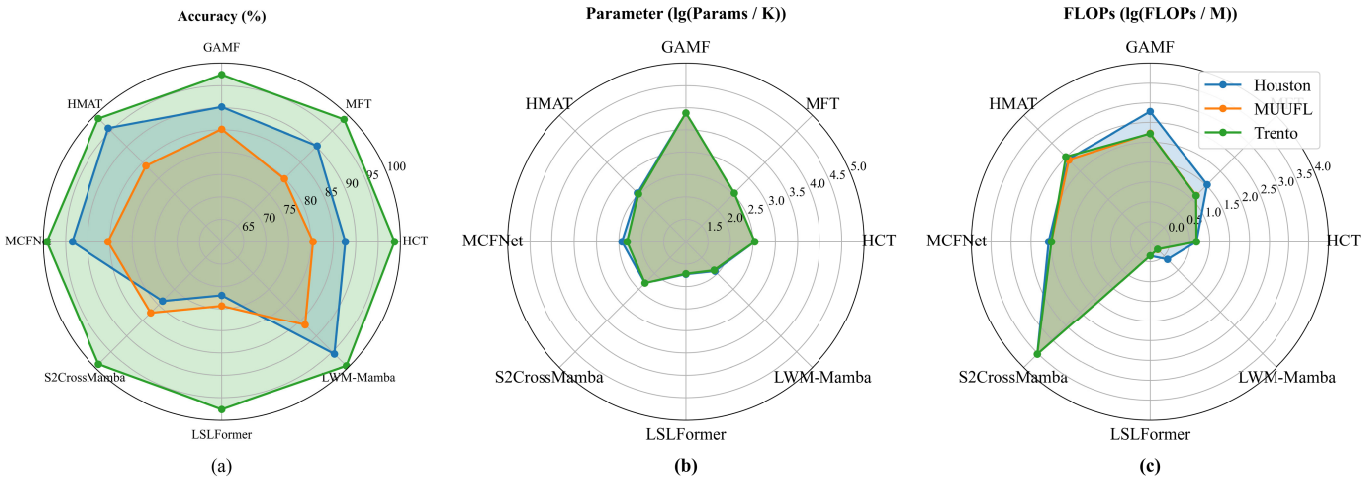


Fig. 13. Comparison of computational complexity of various methods on three datasets. (a) Houston dataset. (b) MUUFL dataset. (c) Trento dataset.

and higher classification fidelity, benefiting from its feature enhancement and multipath scanning mechanisms.

E. Discussion

1) Multipath Scanning Mechanism: To investigate the impact of different scanning paths on classification performance, comparative experiments are conducted for the proposed multipath scanning mechanism, with the scanning strategies illustrated in Fig. 11. Specifically, we compare the two-path row-column scan, the conventional four-directional oriented scan, and the proposed six-path scan with additional center-origin rotation and reverse-rotation paths. As shown in Table VI, the proposed six-path scan achieves the highest OA on Houston, MUUFL, and Trento. Compared with the four-directional oriented scan, it improves OA by 1.55%, 0.75%, and 0.33%, respectively. These results demonstrate that the proposed center-origin rotation and reverse-rotation paths provide complementary spatial context beyond conventional row-column and four-directional scanning. Moreover, the relative performance of the two-path and four-directional variants differs across datasets, indicating that conventional scan orders are sensitive to dataset-specific spatial structures. By integrating conventional directional paths with center-origin paths, the proposed six-path scan achieves more robust spatial context modeling. This finding is also consistent with the motivation of 3DSS-Mamba [38], which shows that scanning routes and dimensional priority can significantly affect Mamba-based HSI representation. Different from 3DSS-Mamba, which investigates five 3-D spectral-spatial scanning routes for single-source HSI tokens, our analysis focuses on multimodal HSI-LiDAR patch classification and studies how different spatial scan-path combinations affect central-pixel prediction. The comparison among the two-path, four-directional, and six-path variants in Table VI justifies the adopted path count in LWM-Mamba and confirms the contribution of the center-origin rotation and reverse-rotation paths.

2) Computational Complexity Analysis: To verify the computational efficiency of the proposed LWM-Mamba module, we conducted a complexity analysis under the Houston dataset setting. For a fair comparison, we kept the overall network architecture unchanged and only replaced the Mamba fusion component with a standard explicit multihead self-attention Transformer block, as shown in Fig. 12. The results show that LWM-Mamba exhibits nearly linear growth in both inference time and memory consumption as the patch size increases, whereas the Transformer counterpart shows much faster memory growth due to the quadratic cost of self-attention. In particular, the Transformer variant becomes infeasible at larger patch sizes, while LWM-Mamba remains scalable. These results confirm that the Mamba-based fusion mechanism effectively reduces the computational burden and is more suitable for large spatial patch modeling in HSI-LiDAR classification tasks.

To further evaluate the lightweight advantage of LWM-Mamba, we compare the parameters and FLOPs of different methods on the Houston, MUUFL, and Trento datasets, as shown in Fig. 13. Overall, LWM-Mamba achieves the highest classification accuracy with relatively low computational cost, demonstrating a favorable balance between efficiency and performance. This advantage mainly benefits from the use of PWConv in the feature embedding module, the efficient design of the frequency enhancement module, and the linear-complexity multipath scanning mechanism. In contrast, methods such as GAMF, MFT, and HCT require substantially higher parameters and FLOPs while yielding lower accuracy. Although MCFNet and S^2CrossMamba reduce computational cost to some extent, their performance remains inferior to LWM-Mamba, especially on challenging datasets such as Houston and MUUFL. These results indicate that LWM-Mamba provides superior efficiency-accuracy trade-offs and achieves state-of-the-art performance with lower computational complexity.

3) Feature Visualization: To visually verify the effectiveness of the MA-Mamba module in enhancing feature discriminability, Fig. 14 presents the t-SNE visualization of feature distributions on the three datasets. From a global perspective, when MA-Mamba is not employed [Fig. 14(a)–(c)], significant overlap between different classes is observed in the

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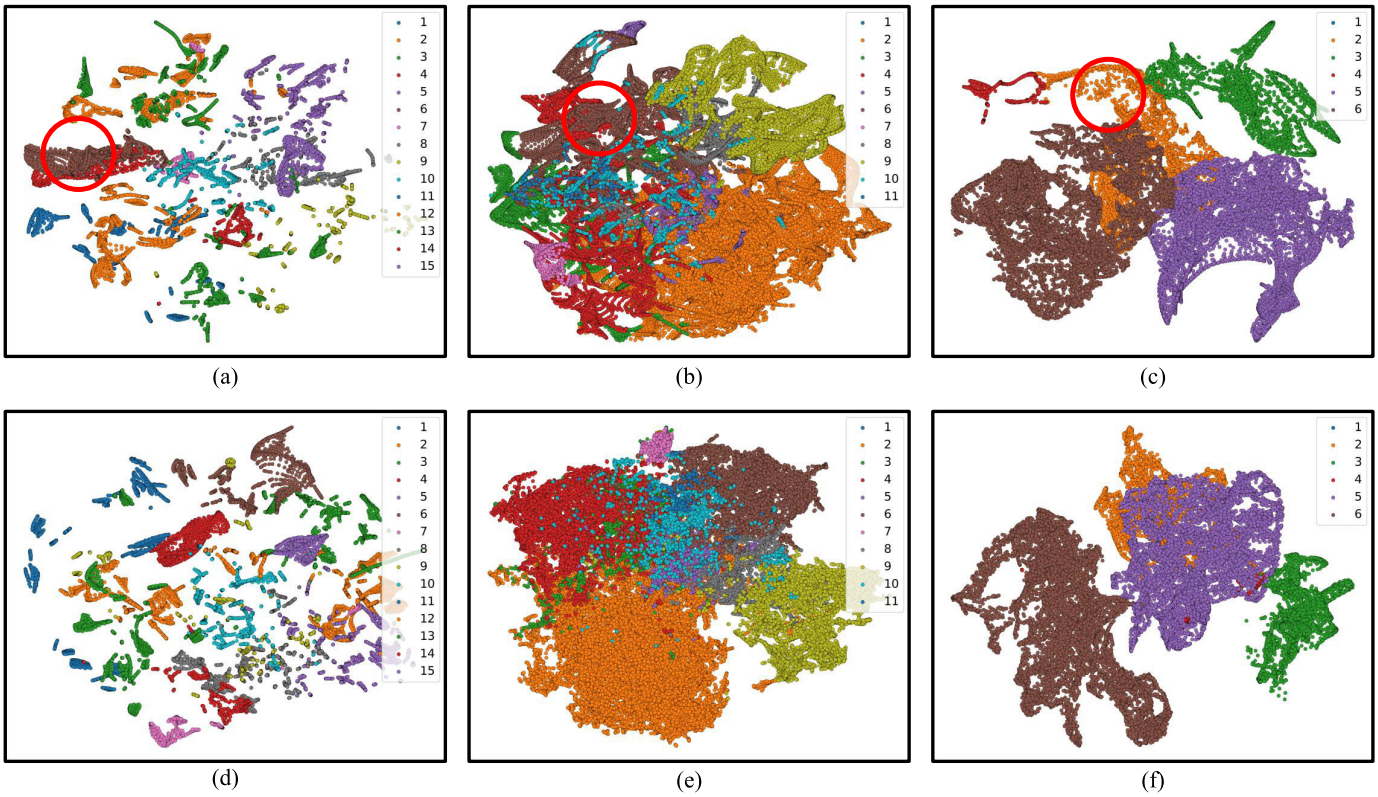


Fig. 14. t-SNE visualization of features with and without the MA-Mamba module. (a) and (d) Houston dataset. (b) and (e) MUUFL dataset. (c) and (f) Trento dataset.

feature space. In particular, land-cover classes with similar spectral characteristics exhibit severe mixing, making them difficult to distinguish. This suggests that although the feature embedding and FEM modules can extract basic features, deep cross-modal fusion remains insufficient, resulting in limited separability in the feature space. After introducing the MA-Mamba module [Fig. 14(d)–(f)], the feature distributions are significantly improved, especially on the MUUFL and Trento datasets. Moreover, the circled regions clearly show that previously mixed samples are pulled back toward their respective class centers. These results further demonstrate that the MA-Mamba module effectively enhances the semantic consistency and class separability of multimodal features.

4) Spatial Neighborhood Size: To investigate the influence of spatial neighborhood size, we further conducted a patch-size sensitivity experiment on the Houston dataset. As shown in Fig. 15, the model achieves the best performance at 11×11. When the patch size is 9×9, the performance is slightly lower, indicating that a smaller neighborhood may provide insufficient spatial contextual information. As the patch size increases beyond 11×11, the classification accuracy gradually decreases. This trend suggests that excessively large patches may introduce redundant background pixels or mixed-class regions, which weakens the discriminative representation of the central pixel. Overall, the results demonstrate that LWM-Mamba is relatively stable under different patch sizes on the Houston dataset, and a moderate patch size provides the best trade-off between spatial context utilization and noise interference.

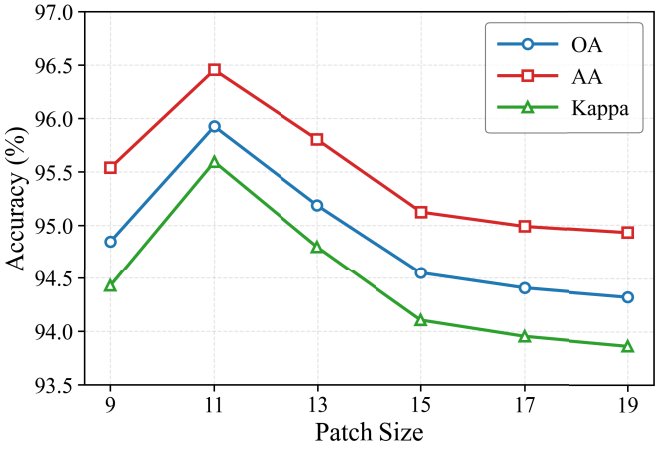


Fig. 15. Impact of different spatial neighborhood sizes on performance on the Houston dataset.

5) Different Training Samples: To evaluate the influence of training sample size, we conducted additional experiments on the Houston dataset by varying the number of training samples per class from 20 to 200, as shown in Fig. 16. However, clear differences can be observed under limited training samples. When only 20 samples per class are available, LWM-Mamba is higher than GAMF, HCT, MFT, MCFNet, and LSLFormer, demonstrating its strong representation capability and robustness in small-sample scenarios. In contrast, LSLFormer shows much lower performance under the same setting, suggesting that it is more sensitive to insufficient

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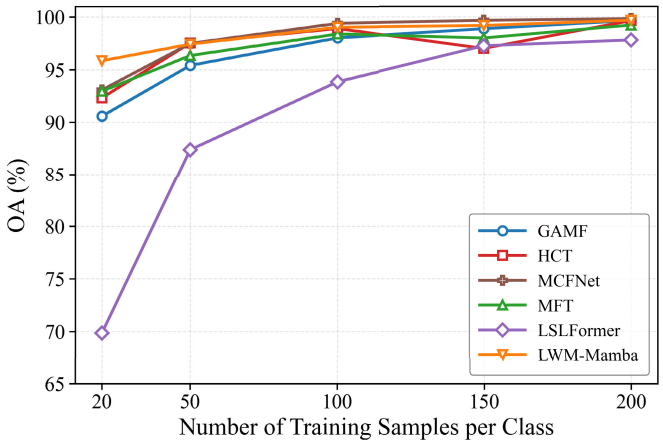


Fig. 16. Impact of different training samples on performance on the Houston dataset.

training samples. Overall, these results indicate that LWM-Mamba can effectively exploit limited labeled samples and further benefit from increased supervision, showing superior robustness and scalability with respect to training sample size on the Houston dataset.

IV. Conclusion

In this article, a LWM-Mamba network, termed LWM-Mamba, is proposed to address the issues of high computational complexity in HSI–LiDAR joint classification and the limited capability of existing Mamba frameworks in modeling spatial context. An efficient feature embedding module is first designed to extract multimodal features with low computational overhead. Then, FEM is introduced to improve feature discriminability through joint frequency-domain and spatial-domain enhancement. The MA-Mamba module incorporates rotation and reverse rotation scanning paths originating from the central pixel, which, together with conventional row–column scanning, enables comprehensive multidirectional coverage and effective modeling of spatial dependencies. Furthermore, an MA-SSM is developed to achieve deep cross-modal interaction. The proposed framework maintains linear computational complexity while achieving a favorable balance between lightweight design and high performance. Experimental results on three public datasets, namely Houston, MUUFL, and Trento, demonstrate that LWM-Mamba significantly outperforms state-of-the-art methods in terms of OA, AA, and Kappa.

Although promising results have been achieved, further improvements are still possible. In future work, additional modalities, such as multispectral and SAR data, will be explored to enhance the completeness of land-cover representation and further improve classification performance.

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